

- 1 (a) Use standard results from the list of formulae (MF19) to show that

$$\sum_{r=1}^n (2-3r)(5-3r) = an^3 + bn^2 + cn,$$

where  $a$ ,  $b$  and  $c$  are integers to be determined.

[3]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

